STA 2002, Summer 2024
Probability and Statistics II
Hypothesis Testing
Part II

Recap: Hypothesis Testing Procedures

1. **Parameter of interest:** From the problem context, identify the parameter of interest
2. **Null hypothesis H_0:** State the null hypothesis – initial belief/claim/assumption
3. **Alternative hypothesis H_1:** Specify an appropriate alternative hypothesis
4. **Test statistic:** Determine an appropriate test statistic
5. **Reject H_0 if :** State the rejection criteria for the null hypothesis
6. **Computation:** Compute any necessary sample quantities, substitute these into the equation for the test statistic, and compute that value
7. **Draw conclusion:** Decide whether or not H_0 should be rejected and report that in the problem context
8. **Report p-values**

- **Three ways to perform hypothesis test**
  - **— Rejection region**
  - **— Compare p-value with α**
  - **— Using confidence interval**

Recap: Errors in Hypothesis Test

Type I Error

Rejecting the null hypothesis H_0 when it is true is defined as a **type I error.**

Probability of Type I Error

$\alpha = P(\text{type I error}) = P(\text{reject } H_0 \text{ when } H_0 \text{ is true})$

Type II Error

Failing to reject the null hypothesis when it is false is defined as a **type II error.**

Probability of Type II Error

$\beta = P(\text{type II error}) = P(\text{fail to reject } H_0 \text{ when } H_0 \text{ is false})$

Decision	H_0 Is True	H_0 Is False
Fail to reject H_0	no error	type II error
Reject H_0	type I error	no error

Recap: Significance Level

- **Generally, the statistician controls type-I error probability α when she selects the critical values**
- **Set type-I error probability at a desired level called significance level**
- **Deal with the corresponding type-II error β resulted from this**

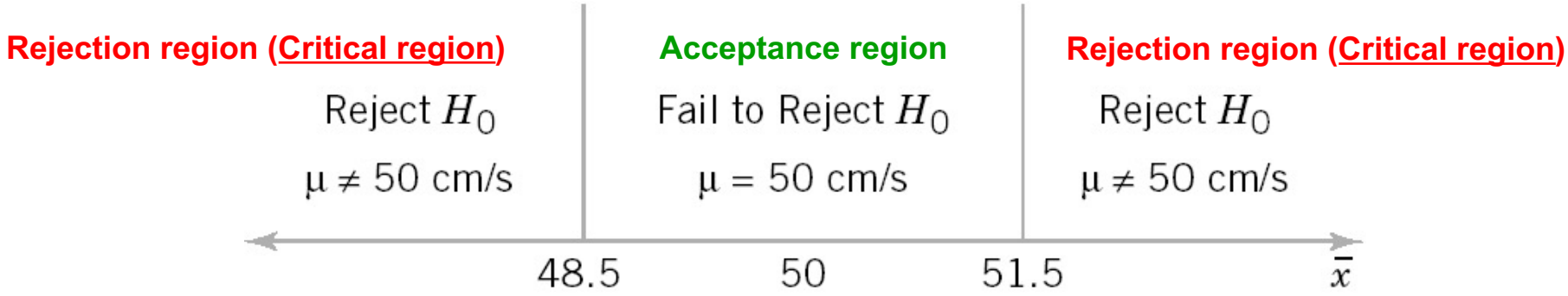
> A widely used procedure in hypothesis testing is to use a type 1 error or significance level of $\alpha = 0.05$. This value has evolved through experience, and may not be appropriate for all situations.

Typical value for significance level:
0.1, 0.05, 0.001

Example

- **E.g. Take 10 samples**

$$H_0: \mu = 50 \text{ centimeters per second}$$
$$H_1: \mu \neq 50 \text{ centimeters per second}$$



Example

Suppose that the standard deviation of burning rate is $\sigma = 2.5$ centimeters per second

$$\alpha = P(\bar{X} < 48.5 \text{ when } \mu = 50) + P(\bar{X} > 51.5 \text{ when } \mu = 50)$$

Type I error

n = 10 samples

$$\sigma/\sqrt{n} = 2.5/\sqrt{10} = 0.79.$$

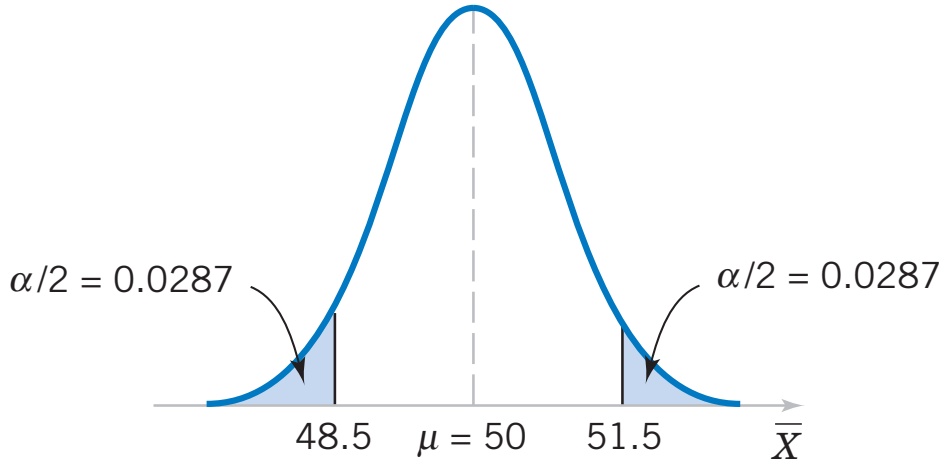


Figure 9-2 The critical region for H_0: $\mu = 50$ versus H_1: $\mu \neq 50$ and $n = 10$.

$$z_1 = \frac{48.5 - 50}{0.79} = -1.90 \quad \text{and} \quad z_2 = \frac{51.5 - 50}{0.79} = 1.90$$

$$\alpha = P(Z < -1.90) + P(Z > 1.90) = 0.0287 + 0.0287 = 0.0574$$

$$\boldsymbol{H_1: \mu = 52}$$

$$\beta = P(48.5 \leq \bar{X} \leq 51.5 \text{ when } \mu = 52)$$

The z-values corresponding to 48.5 and 51.5 when $\mu = 52$ are

$$z_1 = \frac{48.5 - 52}{0.79} = -4.43 \qquad \text{and} \qquad z_2 = \frac{51.5 - 52}{0.79} = -0.63$$

$$\begin{aligned}\beta &= P(-4.43 \leq Z \leq -0.63) = P(Z \leq -0.63) - P(Z \leq -4.43) \\ &= 0.2643 - 0.0000 = 0.2643\end{aligned}$$

Power and Sample Size

Reading: Chapter 8.5 in our textbook.

Statistical Power

$$\underline{\textbf{Power}} = 1 - \beta = P(\text{reject } H_0 \text{ when } H_1 \text{ is true})$$

Power

The **power** of a statistical test is the probability of rejecting the null hypothesis H_0 when the alternative hypothesis is true.

Statistical Power

Null Hypothesis	Alternative Hypothesis	Test Statistic	Critical region
$H_0 : \mu = \mu_0$	$H_1 : \mu \neq \mu_0$	T	**C**
	or $H_1 : \mu > \mu_0$		
	or $H_1 : \mu < \mu_0$		

Test: reject the null hypothesis if $T \in C$.

$$\alpha = P(\text{Test makes Type I error}) = P(T \in C; H_0) = P(T \in C; \mu = \mu_0)$$

$$\beta(\mu_1) = P(\text{Test makes Type II error under } \mu = \mu_1) = P(T \notin C; \mu = \mu_1)$$

$$\text{Power } K(\mu_1) = 1 - \beta(\mu_1) = P(T \in C; \mu = \mu_1)$$

$$K(\mu_0) = \mathrm{P}(T \in C; \mu_0) = \alpha$$

Type I error

Power for testing normal mean

Testing normal mean
(Known Variance, two-sided)

Data samples

$$X_1, \ldots, X_n \sim N(\mu, \sigma^2)$$

Null Hypothesis

$$H_0 : \mu = \mu_0$$

Alternative Hypothesis

$$H_1 : \mu \neq \mu_0$$

Test Statistic

$$Z_0 = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$$

Critical region

C: $|Z_0| > Z_{\alpha/2}$

Distribution under H_0

$$Z_0 \sim N(0,1)$$

Power function $\boldsymbol{K(\mu_1) = 1 - \beta = P(Z_0 \in C; \mu_1) = P(|Z_0| > Z_{\alpha/2}; \mu_1)}$

Power function, denoted by $K(\mu)$, is a function of the parameter μ.

Testing normal mean
(Known Variance, two-sided)

Calculation of the power function:

$$K(\mu_1) = \mathrm{P}\left(|Z_0| > Z_{\alpha/2}; \mu_1\right)$$

$$= \mathrm{P}\left(Z_0 > Z_{\alpha/2}; \mu_1\right) + \mathrm{P}\left(Z_0 < -Z_{\alpha/2}; \mu_1\right)$$

$$= \mathrm{P}(Z_0 - \frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}} > Z_{\alpha/2} - \frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}})$$

$$+ \mathrm{P}(Z_0 - \frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}} < -Z_{\alpha/2} - \frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}})$$

$$= \Phi\left(-Z_{\alpha/2} + \frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}}\right) + \Phi\left(-Z_{\alpha/2} - \frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}}\right)$$

Under $\mu = \mu_1$

$$X_1, \ldots, X_n \sim N(\mu_1, \sigma^2)$$

$$Z_0 = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(\frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}}, 1)$$

Testing normal mean
(Known Variance, one-sided)

Data samples

$$X_1, \dots, X_n \sim N(\mu, \sigma^2)$$

Null Hypothesis

$$H_0 : \mu = \mu_0$$

Alternative Hypothesis

$$H_1 : \mu > \mu_0$$

Test Statistic

$$Z_0 = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

Critical region

$$\mathbf{C:}\ Z_0 > Z_\alpha$$

Distribution under H_0

$$Z_0 \sim N(0,1)$$

Under $\mu = \mu_1 > \mu_0$

Power function $\boldsymbol{K(\mu_1) = 1 - \beta = P(Z_0 \in C; \mu_1) = P(Z_0 > Z_\alpha; \mu_1)}$

Under $\mu = \mu_1$

$$Z_0 = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N\left(\frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}}, 1\right)$$

$$= P\left(Z_0 - \frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}} > Z_\alpha - \frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}}\right)$$

$$= \Phi\left(-Z_\alpha + \frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}}\right)$$

Testing normal mean
(Known Variance, one-sided)

Data samples

$$X_1, \ldots, X_n \sim N(\mu, \sigma^2)$$

Null Hypothesis

$$H_0 : \mu = \mu_0$$

Alternative Hypothesis

$$H_1 : \mu < \mu_0$$

Test Statistic

$$Z_0 = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

Critical region

$$\mathbf{C:}\ Z_0 < -Z_\alpha$$

Distribution under H_0

$$Z_0 \sim N(0,1)$$

Under $\mu = \mu_1 < \mu_0$

Power function $K(\mu_1) = 1 - \beta = P(Z_0 \in C; \mu_1) = P(Z_0 < -Z_\alpha; \mu_1)$

$$= P(Z_0 - \frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}} < -Z_\alpha - \frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}})$$

$$= \Phi\left(-Z_\alpha + \frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}}\right)$$

Under $\mu = \mu_1$

$$Z_0 = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(\frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}}, 1)$$

Testing normal mean

Factors Affecting the Power

(1) If the significance level is made smaller (α decreases), z_α increases and hence the power decreases.

(2) If the alternative mean is shifted farther away from the null mean ($|\mu_0 - \mu_1|$ increases), then the power increases.

(3) If the standard deviation of the distribution of individual observations increases (σ increases), then the power decreases.

(4) If the sample size increases (n increases), then the power increases.

$$\Phi\left(-Z_{\alpha/2} + \frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}}\right) + \Phi\left(-Z_{\alpha/2} - \frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}}\right)$$

two-sided

$$\Phi\left(-Z_{\alpha} + \frac{|\mu_0 - \mu_1|}{\sigma/\sqrt{n}}\right)$$

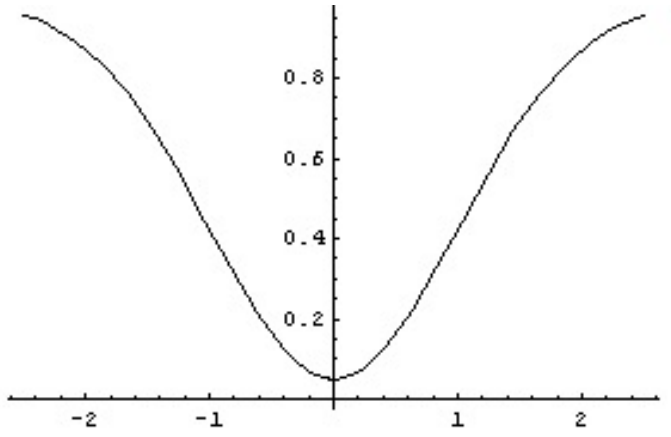
one-sided

Example:
power function for the test

$$H_0 : \mu = 0$$

$$H_1 : \mu \neq 0$$

P(reject H_0)



Example

Suppose $X_1, \ldots, X_{25}$ are the exam scores of STA2002 of $n = 25$ students. We assume that they are i.i.d. normally distributed $N(\mu, 100)$.

We want to test

$$H_0 : \mu = 60, \quad H_1 : \mu > 60.$$

We use the test Statistic $Z_0 = \dfrac{\overline{X} - \mu_0}{\sigma / \sqrt{n}}$

Critical region: reject null hypothesis H_0 if $\boldsymbol{Z_0 > Z_\alpha}$

Q1: What is the power function $K(\mu)$ of this test?

Q2: Suppose the true mean is 65, what is the power of this test?

Q3: Plot the function $K(\mu)$.

Example

At significance level α:

critical region $z_0 > z_\alpha$.

(1) power function

$$K(\mu_1) = P(z_0 > z_\alpha \mid \mu = \mu_1)$$

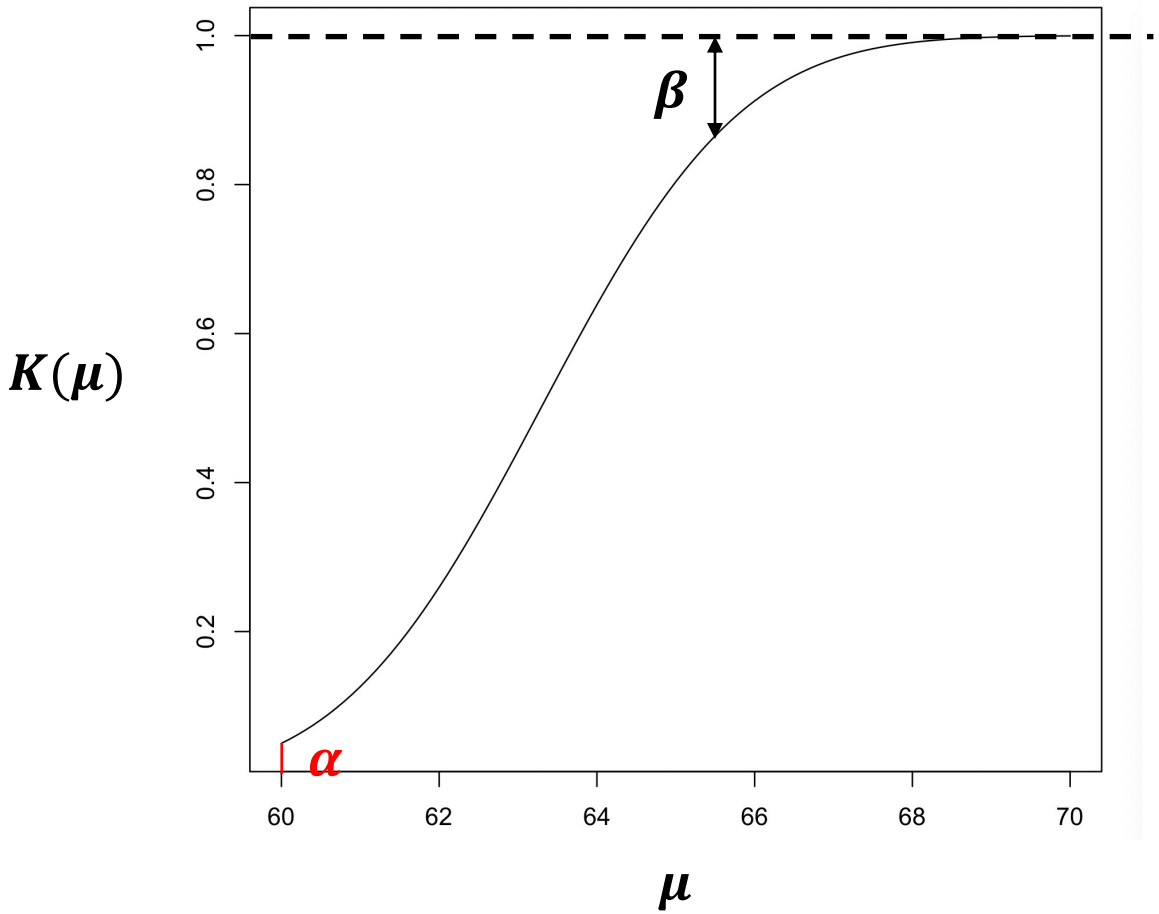
$$= \Phi\left(-z_\alpha + \frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}}\right)$$

$$= \Phi\left(-1.645 + \frac{\mu_1 - 60}{10/\sqrt{25}}\right)$$

$$= \Phi\left(-1.645 + \frac{\mu_1 - 60}{2}\right)$$

(2) when $\mu = \mu_1 = 65$, the power

$$K(65) = \Phi\left(-1.645 + \frac{5}{2}\right) = \Phi(0.855) = 0.803$$



Sample Size: Testing normal mean
(Known Variance, one-sided)

A one-sided test will be conducted at level α
The true alternative mean is assumed to be μ_1

$$H_0 : \mu = \mu_0 \qquad H_1 : \mu > \mu_0$$
$$H_1 : \mu < \mu_0$$

What sample size is needed to be able to detect a significant difference (reject H_0) with probability 1-β ?

The power function:

$$K(\mu_1) = \Phi\left(-Z_\alpha + \frac{|\mu_0 - \mu_1|}{\sigma/\sqrt{n}}\right) = 1 - \beta$$

$$\Rightarrow \quad -Z_\alpha + \frac{|\mu_1 - \mu_0|}{\sigma/\sqrt{n}} = Z_\beta \quad \Rightarrow \quad \boxed{n = \frac{(Z_\beta + Z_\alpha)^2 \sigma^2}{(\mu_1 - \mu_0)^2}}$$

Sample Size: Testing normal mean
(Known Variance, two-sided)

A two-sided test will be conducted at level α
The true alternative mean is assumed to be μ_1

$$H_0 : \mu = \mu_0 \qquad H_1 : \mu \neq \mu_0$$

What sample size is needed to be able to detect a significant difference (reject H_0) with probability 1-β ?

The power function:

$$K(\mu_1) = \mathrm{P}\left(|Z_0| > Z_{\alpha/2}; \mu_1\right) = \mathrm{P}\left(Z_0 > Z_{\alpha/2}; \mu_1\right) + \mathrm{P}\left(Z_0 < -Z_{\alpha/2}; \mu_1\right)$$

$$= \boxed{\Phi\left(-Z_{\alpha/2} + \frac{\mu_1-\mu_0}{\sigma/\sqrt{n}}\right) + \Phi\left(-Z_{\alpha/2} - \frac{\mu_1-\mu_0}{\sigma/\sqrt{n}}\right)} = 1-\beta$$

$$\Rightarrow \quad -Z_{\alpha/2} + \frac{|\mu_0-\mu_1|}{\sigma/\sqrt{n}} \approx Z_\beta \quad \Rightarrow \quad \boxed{n \approx \frac{\left(Z_\beta + Z_{\alpha/2}\right)^2 \sigma^2}{(\mu_1-\mu_0)^2}}$$

Using only the dominant term (the other is pretty much zero)

Testing normal mean

Null Hypothesis	Test Statistic	Distribution under H_0
$H_0 : \mu = \mu_0$	$Z_0 = \dfrac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$	$Z_0 \sim N(0,1)$

Alternative Hypothesis	Critical Region (H_0 is rejected)	Power under $\mu = \mu_1$	Sample size needed for $K(\mu_1) = 1 - \beta; K(\mu_0) = \alpha$
$H_1 : \mu \neq \mu_0$	$\lvert Z_0 \rvert > Z_{\alpha/2}$	$\Phi\left(-Z_{\alpha/2} + \frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}}\right) + \Phi\left(-Z_{\alpha/2} - \frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}}\right)$	$n = \dfrac{\left(Z_\beta + Z_{\alpha/2}\right)^2 \sigma^2}{(\mu_1 - \mu_0)^2}$
$H_1 : \mu > \mu_0$	$Z_0 > Z_\alpha$	$\Phi\left(-Z_\alpha + \dfrac{\lvert\mu_0 - \mu_1\rvert}{\sigma/\sqrt{n}}\right)$	$n = \dfrac{\left(Z_\beta + Z_\alpha\right)^2 \sigma^2}{(\mu_1 - \mu_0)^2}$
$H_1 : \mu < \mu_0$	$Z_0 < -Z_\alpha$		

Sample size

Factors Affecting the Sample Size

(1) The sample size increases as σ^2 increases.

(2) The sample size increases as the significance level is made smaller (α decreases).

(3) The sample size increases as the required power increases ($1 - \beta$ increases).

(4) The sample size decreases as the absolute value of the distance between the null and alternative means ($|\mu_0 - \mu_1|$) increases.

$$n = \frac{\left(Z_\beta + Z_{\alpha/2}\right)^2 \sigma^2}{(\mu_1 - \mu_0)^2}$$

two-sided

$$n = \frac{\left(Z_\beta + Z_\alpha\right)^2 \sigma^2}{(\mu_1 - \mu_0)^2}$$

one-sided

Example

Suppose $X_1, \ldots, X_{25}$ are the exam scores of STA2002 of $n = 25$ students. We assume that they are i.i.d. normally distributed $N(\mu, 100)$.

We want to test

$$H_0 : \mu = 60, \quad H_1 : \mu > 60.$$

Q4: Suppose the true mean is 65, what is the sample size needed to have power at least 0.95?

(4) sample size $1-\beta = 0.95$ $\beta = 0.05$

$$n = \frac{(z_\alpha + z_\beta)^2 \sigma^2}{(\mu_1 - \mu_0)^2} = \frac{(z_{0.05} + z_{0.05})^2 \cdot \sigma^2}{(65-60)^2}$$

$$= \frac{(1.645 + 1.645)^2 \times 100}{5^2}$$

$$= 43.2964$$

$$\Rightarrow \underline{n = 44}$$

Power for two-samples

Testing Equality of Two Normal Means (known variances, two-sided)

Data samples

$$X_1, \dots, X_n \sim N(\mu_X, \sigma_X^2)$$

$$Y_1, \dots, Y_m \sim N(\mu_Y, \sigma_Y^2)$$

Null Hypothesis

$$H_0: \mu_X = \mu_Y$$

Alternative Hypothesis

$$H_1: \mu_X \neq \mu_Y$$

Test Statistic

$$Z_0 = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}}$$

Critical region

$$\mathbf{C:}\ |Z_0| > Z_{\alpha/2}$$

Distribution under H_0

$$Z_0 \sim N(0,1)$$

Power function $\boldsymbol{K(\Delta) = 1 - \beta = P(Z_0 \in C; \Delta) = P(|Z_0| > Z_{\alpha/2}; \Delta)}$

$$\boldsymbol{\mu_X - \mu_Y = \Delta}$$

Testing Equality of Two Normal Means (known variances, two-sided)

Calculation of the power function:

$$K(\Delta) = \mathrm{P}(|Z_0| > Z_{\alpha/2}; \Delta)$$

$$= \mathrm{P}\left(Z_0 - \frac{\Delta}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}} > Z_{\alpha/2} - \frac{\Delta}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}}\right)$$

$$+ \mathrm{P}\left(Z_0 - \frac{\Delta}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}} < -Z_{\alpha/2} - \frac{\Delta}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}}\right)$$

$$= \Phi\left(-Z_{\alpha/2} + \frac{\Delta}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}}\right) + \Phi\left(-Z_{\alpha/2} - \frac{\Delta}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}}\right)$$

Under $\mu_X - \mu_Y = \Delta$

$$Z_0 = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}} \sim N\left(\frac{\Delta}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}}, 1\right)$$

Under $n = m, \sigma_X = \sigma_Y = \sigma$, we have

$$K(\Delta) = \Phi\left(-Z_{\alpha/2} + \frac{\Delta}{\sigma}\sqrt{\frac{n}{2}}\right) + \Phi\left(-Z_{\alpha/2} - \frac{\Delta}{\sigma}\sqrt{\frac{n}{2}}\right)$$

Testing Equality of Two Normal Means (known variances, one-sided)

Data samples

$$X_1, \dots, X_n \sim N(\mu_X, \sigma_X^2)$$

$$Y_1, \dots, Y_m \sim N(\mu_Y, \sigma_Y^2)$$

Null Hypothesis

$$H_0: \mu_X = \mu_Y$$

Alternative Hypothesis

$$H_1: \mu_X > \mu_Y$$

Test Statistic

$$Z_0 = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}}$$

Critical region

C: $Z_0 > Z_\alpha$

Distribution under H_0

$$Z_0 \sim N(0,1)$$

Under $\mu_X - \mu_Y = \Delta > 0$

Power function $K(\Delta) = 1 - \beta = P(Z_0 \in C; \Delta) = P(Z_0 > Z_\alpha; \Delta)$

$$= \Phi\left(-Z_\alpha + \frac{|\Delta|}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}}\right) \xrightarrow{n=m, \sigma_X=\sigma_Y=\sigma} \Phi\left(-Z_\alpha + \frac{|\Delta|}{\sigma}\sqrt{\frac{n}{2}}\right)$$

Testing Equality of Two Normal Means (known variances, one-sided)

Data samples

$X_1, \dots, X_n \sim N(\mu_X, \sigma_X^2)$

$Y_1, \dots, Y_m \sim N(\mu_Y, \sigma_Y^2)$

Null Hypothesis

$H_0: \mu_X = \mu_Y$

Alternative Hypothesis

$H_1: \mu_X < \mu_Y$

Test Statistic

$$Z_0 = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}}$$

Critical region

C: $Z_0 < -Z_\alpha$

Distribution under H_0

$Z_0 \sim N(0,1)$

Under $\mu_X - \mu_Y = \Delta < 0$

Power function $K(\Delta) = 1 - \beta = P(Z_0 \in C; \Delta) = P(Z_0 < -Z_\alpha; \Delta)$

$$= \Phi\left(-Z_\alpha + \frac{|\Delta|}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}}\right) \xrightarrow{n=m, \sigma_X=\sigma_Y=\sigma} \Phi\left(-Z_\alpha + \frac{|\Delta|}{\sigma}\sqrt{\frac{n}{2}}\right)$$

Sample Size: Testing Equality of Two Normal Means

(Known Variance, two-sided)

Under $n = m, \sigma_X = \sigma_Y = \sigma$

$$H_0: \mu_X = \mu_Y \qquad H_1: \mu_X \neq \mu_Y$$

A two-sided test will be conducted at level α
The true alternative mean is assumed to be μ_1

What sample size is needed to be able to detect a significant difference (reject H_0) with probability 1-β ?

The power function:

$$K(\Delta) = \Phi\left(-Z_{\alpha/2} + \frac{\Delta}{\sigma}\sqrt{\frac{n}{2}}\right) + \Phi\left(-Z_{\alpha/2} - \frac{\Delta}{\sigma}\sqrt{\frac{n}{2}}\right) = 1 - \beta$$

$$\Rightarrow \quad -Z_{\alpha/2} + \frac{|\Delta|}{\sigma}\sqrt{\frac{n}{2}} \approx Z_\beta \quad \Rightarrow \quad \boxed{n = 2\frac{\left(Z_\beta + Z_{\alpha/2}\right)^2 \sigma^2}{\Delta^2}} \qquad \mu_X - \mu_Y = \Delta$$

Using only the dominant term (the other is pretty much zero)

Sample Size: Testing Equality of Two Normal Means

(Known Variance, one-sided)

Under $n = m, \sigma_X = \sigma_Y = \sigma$

$$H_0: \mu_X = \mu_Y \qquad H_1: \mu_X > \mu_Y$$

$$H_1: \mu_X < \mu_Y$$

A one-sided test will be conducted at level α
The true alternative mean is assumed to be Δ

What sample size is needed to be able to detect a significant difference (reject H_0) with probability 1-β ?

The power function for one-sided test:

$$K(\Delta) = \Phi\left(-Z_\alpha + \frac{|\Delta|}{\sigma}\sqrt{\frac{n}{2}}\right) = 1 - \beta$$

$$\Rightarrow \quad -Z_\alpha + \frac{|\Delta|}{\sigma}\sqrt{\frac{n}{2}} \approx Z_\beta \quad \Rightarrow \quad \boxed{n = 2\frac{\left(Z_\beta + Z_\alpha\right)^2 \sigma^2}{\Delta^2}} \qquad \boldsymbol{\mu_X - \mu_Y = \Delta}$$

Using only the dominant term (the other is pretty much zero)

Testing Equality of Two Normal Means

$n = m$,
$\sigma_X = \sigma_Y = \sigma$,

Null Hypothesis

$$H_0: \mu_X = \mu_Y$$

Test Statistic

$$Z_0 = \frac{\bar{X} - \bar{Y}}{\sigma\sqrt{2/n}}$$

Distribution under H_0

$$Z_0 \sim N(0,1)$$

Alternative Hypothesis	Critical Region (H_0 is rejected)	Power under $\mu_X - \mu_Y = \Delta$	Sample size needed for $K(\Delta) = 1 - \beta; K(0) = \alpha$
$H_1: \mu_X \neq \mu_Y$	$\lvert Z_0 \rvert > Z_{\alpha/2}$	$\Phi\left(-Z_{\alpha/2} + \frac{\Delta}{\sigma}\sqrt{\frac{n}{2}}\right) + \Phi\left(-Z_{\alpha/2} - \frac{\Delta}{\sigma}\sqrt{\frac{n}{2}}\right)$	$n = 2\frac{\left(Z_\beta + Z_{\alpha/2}\right)^2 \sigma^2}{\Delta^2}$
$H_1: \mu_X > \mu_Y$	$Z_0 > Z_\alpha$	$\Phi\left(-Z_\alpha + \frac{\lvert\Delta\rvert}{\sigma}\sqrt{\frac{n}{2}}\right)$	$n = 2\frac{\left(Z_\beta + Z_\alpha\right)^2 \sigma^2}{\Delta^2}$
$H_1: \mu_X < \mu_Y$	$Z_0 < -Z_\alpha$		

Example

Let X be the driving speed of male, and Y be the driving speed of female. Assume they are independent normal. Assume that driving speed of both female and male follow normal distributions with a common variance 25. We would like to test

$$H_0: \mu_X = \mu_Y$$

$$H_1: \mu_X > \mu_Y$$

Data	Male (X)	Female (Y)
	$n = 29$	$n = 29$
	$\bar{X} = 104.1$	$\bar{Y} = 90.9$

Q1: Suppose the true mean satisfy $\mu_X - \mu_Y = 5$, what is the power of this test?

Q2: When $\mu_X - \mu_Y = 5$, what is the sample size needed in order to have power at least 0.9?

Q1. $\Delta = \mu_X - \mu_Y = 5$, $n = 29$. $\sigma^2 = 25$

power $K(\Delta) = \Phi\left(-z_\alpha + \frac{|\Delta|}{\sigma}\sqrt{\frac{n}{2}}\right)$

$K(5) = \Phi\left(-1.645 + \frac{5}{5} \times \sqrt{\frac{29}{2}}\right)$

$= \Phi(2.16) = 0.9846$

Q2. $1 - \beta = 0.9 \Rightarrow \beta = 0.1$.

$z_\beta = z_{0.1} = 1.28$

$$n = 2\frac{(z_\alpha + z_\beta)^2\sigma^2}{\Delta^2}$$

$$= 2 \times \frac{(1.28 + 1.645)^2 \times 25}{25}$$

$= 17.11$

$\Rightarrow n = 18$

Power for proportions

Testing Proportion
(two-sided)

Data samples

$$X_1, X_2, \cdots, X_n \sim Bernoulli(p)$$

Null Hypothesis

$$H_0 : p = p_0$$

Alternative Hypothesis

$$H_1 : p \neq p_0$$

Test Statistic

$$Z_0 = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

Critical region

$$\mathbf{C:}\ |Z_0| > Z_{\alpha/2}$$

Distribution under H_0

$$Z_0 \sim N(0,1)$$

Power function $\boldsymbol{K(p_1) = 1 - \beta = P(Z_0 \in C; p_1) = P(|Z_0| > Z_{\alpha/2}; p_1)}$

Under $\boldsymbol{p = p_1}$

Testing Proportion
(two-sided)

Calculation of the power function:

$$K(p_1) = \mathrm{P}\left(|Z_0| > Z_{\alpha/2}; p_1\right)$$

$$= \mathrm{P}\left(Z_0 > Z_{\alpha/2}; p_1\right) + \mathrm{P}\left(Z_0 < -Z_{\alpha/2}; p_1\right)$$

$$= \mathrm{P}\left(\frac{Z_0 - \frac{p_1 - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}}{\sqrt{\frac{p_1(1-p_1)}{p_0(1-p_0)}}} > \frac{Z_{\alpha/2} - \frac{p_1 - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}}{\sqrt{\frac{p_1(1-p_1)}{p_0(1-p_0)}}}\right)$$

$$+ \mathrm{P}\left(\frac{Z_0 - \frac{p_1 - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}}{\sqrt{\frac{p_1(1-p_1)}{p_0(1-p_0)}}} < \frac{-Z_{\alpha/2} - \frac{p_1 - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}}{\sqrt{\frac{p_1(1-p_1)}{p_0(1-p_0)}}}\right)$$

Under $p = p_1$

$$\frac{\hat{p} - p_1}{\sqrt{\frac{p_1(1-p_1)}{n}}} \sim N(0,1)$$

$$Z_0 = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} \sim N\left(\frac{p_1 - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}, \frac{p_1(1-p_1)}{p_0(1-p_0)}\right)$$

$$= \Phi\left(\frac{-Z_{\alpha/2}\sqrt{p_0(1-p_0)} + \sqrt{n}(p_1 - p_0)}{\sqrt{p_1(1-p_1)}}\right) + \Phi\left(\frac{-Z_{\alpha/2}\sqrt{p_0(1-p_0)} - \sqrt{n}(p_1 - p_0)}{\sqrt{p_1(1-p_1)}}\right)$$

Testing Proportion
(one-sided)

Data samples

$$X_1, X_2, \cdots, X_n \sim Bernoulli(p)$$

Null Hypothesis

$$H_0 : p = p_0$$

Alternative Hypothesis

$$H_1 : p > p_0$$

Test Statistic

$$Z_0 = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

Critical region

C: $Z_0 > Z_\alpha$

Distribution under H_0

$$Z_0 \sim N(0,1)$$

Under $p = p_1 > p_0$

Power function $K(p_1) = 1 - \beta = P(Z_0 \in C; p_1) = P(Z_0 > Z_\alpha; p_1)$

$$= \Phi\left(\frac{-Z_\alpha\sqrt{p_0(1-p_0)} + \sqrt{n}|p_1 - p_0|}{\sqrt{p_1(1-p_1)}}\right)$$

Testing Proportion
(one-sided)

Data samples

$$X_1, X_2, \cdots, X_n \sim Bernoulli(p)$$

Null Hypothesis

$$H_0 : p = p_0$$

Alternative Hypothesis

$$H_1 : p < p_0$$

Test Statistic

$$Z_0 = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

Critical region

C: $Z_0 < -Z_\alpha$

Distribution under H_0

$$Z_0 \sim N(0,1)$$

Under $p = p_1 < p_0$

Power function $K(p_1) = 1 - \beta = P(Z_0 \in C; p_1) = P(Z_0 < -Z_\alpha; p_1)$

$$= \Phi\left(\frac{-Z_\alpha\sqrt{p_0(1-p_0)} + \sqrt{n}|p_1 - p_0|}{\sqrt{p_1(1-p_1)}}\right)$$

Sample size: Testing Proportion
(two-sided)

A two-sided test will be conducted at level α
The true alternative mean is assumed to be p_1

$$H_0 : p = p_0 \qquad H_1 : p \neq p_0$$

What sample size is needed to be able to detect a significant difference (reject H_0) with probability 1-β ?

The power function:

$$K(p_1) = \Phi\left(\frac{-Z_{\alpha/2}\sqrt{p_0(1-p_0)} + \sqrt{n}(p_1 - p_0)}{\sqrt{p_1(1-p_1)}}\right) + \Phi\left(\frac{-Z_{\alpha/2}\sqrt{p_0(1-p_0)} - \sqrt{n}(p_1 - p_0)}{\sqrt{p_1(1-p_1)}}\right) = 1 - \beta$$

$$\frac{-Z_{\alpha/2}\sqrt{p_0(1-p_0)} + \sqrt{n}|p_1 - p_0|}{\sqrt{p_1(1-p_1)}} \approx Z_\beta$$

Using only the dominant term (the other is pretty much zero)

$$n = \frac{\left(Z_\beta\sqrt{p_1(1-p_1)} + Z_{\alpha/2}\sqrt{p_0(1-p_0)}\right)^2}{(p_1 - p_0)^2}$$

Sample size: Testing Proportion
(one-sided)

A one-sided test will be conducted at level α
The true alternative mean is assumed to be p_1

$H_0 : p = p_0$

$H_1 : p > p_0$

$H_1 : p < p_0$

What sample size is needed to be able to detect a significant difference (reject H_0) with probability 1-β ?

The power function:

$$K(p_1) = \Phi\left(\frac{-Z_\alpha\sqrt{p_0(1-p_0)} + \sqrt{n}|p_1 - p_0|}{\sqrt{p_1(1-p_1)}}\right) = 1 - \beta$$

$$\frac{-Z_\alpha\sqrt{p_0(1-p_0)} + \sqrt{n}|p_1 - p_0|}{\sqrt{p_1(1-p_1)}} = Z_\beta$$

$$n = \frac{\left(Z_\beta\sqrt{p_1(1-p_1)} + Z_\alpha\sqrt{p_0(1-p_0)}\right)^2}{(p_1 - p_0)^2}$$

Testing Proportion

Null Hypothesis	Test Statistic	Distribution under H_0
$H_0 : p = p_0$	$Z_0 = \dfrac{\hat{p} - p_0}{\sqrt{\dfrac{p_0(1-p_0)}{n}}}$	$Z_0 \sim N(0,1)$

Alternative Hypothesis	Critical Region (H_0 is rejected)	Power under $p = p_1$	Sample size needed for $K(\Delta) = 1-\beta; K(0) = \alpha$
$H_1 : p \neq p_0$	$\lvert Z_0 \rvert > Z_{\alpha/2}$	$\Phi\left(\frac{-Z_{\alpha/2}\sqrt{p_0(1-p_0)} + \sqrt{n}(p_1-p_0)}{\sqrt{p_1(1-p_1)}}\right) + \Phi\left(\frac{-Z_{\alpha/2}\sqrt{p_0(1-p_0)} - \sqrt{n}(p_1-p_0)}{\sqrt{p_1(1-p_1)}}\right)$	$n = \dfrac{\left(Z_\beta\sqrt{p_1(1-p_1)} + Z_{\alpha/2}\sqrt{p_0(1-p_0)}\right)^2}{(p_1-p_0)^2}$
$H_1 : p > p_0$	$Z_0 > Z_\alpha$	$\Phi\left(\dfrac{-Z_\alpha\sqrt{p_0(1-p_0)} + \sqrt{n}\lvert p_1-p_0\rvert}{\sqrt{p_1(1-p_1)}}\right)$	$n = \dfrac{\left(Z_\beta\sqrt{p_1(1-p_1)} + Z_\alpha\sqrt{p_0(1-p_0)}\right)^2}{(p_1-p_0)^2}$
$H_1 : p < p_0$	$Z_0 < -Z_\alpha$		

Example

Suppose $X_1, \ldots, X_{20}$ are i.i.d. Bernoulli trials with success probability p. To test

$$H_0 : p = \frac{1}{2}, \quad H_1 : p < \frac{1}{2},$$

We use the test Statistic $$Z_0 = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

Critical region: reject null hypothesis H_0 if $Z_0 < -Z_\alpha$

Q1: What is the power function $K(p)$ of this test?
Q2: Suppose the true proportion is 0.2, what is the power of this test?
Q3: For a true proportion 0.4, what is the sample size needed to have power 0.8?

1. power $k(p_1) = \Phi\left(\frac{-z_\alpha \sqrt{p_0(1-p_0)} + \sqrt{n}\,|p_1 - p_0|}{\sqrt{p_1(1-p_1)}} \right)$

2. let $p_1 = 0.2$

$$\Rightarrow k(0.2) = \Phi\left(\frac{-1.645 \times \sqrt{\frac{1}{2} \times \frac{1}{2}} + \sqrt{20} \times |0.2 - 0.5|}{\sqrt{0.2 \times 0.8}} \right)$$

$$= \Phi(1.2979) = 0.90$$

3. power $= 1 - \beta = 0.8$ $\quad \beta = 0.2$

$z_\beta = 0.84$

$$n = \frac{\left[z_\alpha \sqrt{p_0(1-p_0)} + z_\beta \sqrt{p_1(1-p_1)} \right]^2}{(p_1 - p_0)^2}$$

$$= \frac{\left(1.645 \times \frac{1}{2} + 0.84 \times \sqrt{0.4 \times 0.6} \right)^2}{(0.4 - 0.5)^2}$$

$$= 152.28$$

$$n = 153$$